ELEMENTARY CERTIFIED FACTS FOR THE FERMAT QUINTIC FOURFOLD $V(F_5) \subset \mathbb{P}^5$: THE EULER IDENTITY, AN EULER DEFECT, AND THE ISOLATED INTEGER CRITICAL POINT OF THE AFFINE CONE

BENJAMIN STANLEY FROHMAN

Abstract. We formalize in Lean 4, using the core library only, a small set of elementary facts about the Fermat quintic polynomial $F_5 = \sum_{i=0}^{5} x_i^5$ evaluated on integers: Euler's identity $5F_5 = \sum x_i\, \partial_i F_5$; the exact Euler defect $-5\prod x_i$ of the non-homogeneous perturbation $G = F_5 - 5\prod x_i$; and the fact that the only integer point at which all six partial derivatives $5x_i^4$ vanish is the origin. We also record, as checked arithmetic and string labels, the bookkeeping constants $126 - 6 = 120$ and 1751 that appear in Hodge-theoretic discussions of this host, and we state explicitly that no cohomology group, Hodge number, or algebraic cycle is constructed. All proofs are free of `sorry` and use only Lean's standard axioms.

1. Introduction

The Fermat quintic fourfold $V(F_5) = \{x_0^5 + \cdots + x_5^5 = 0\} \subset \mathbb{P}^5$ is a standard smooth test case for Hodge-theoretic computations. Smoothness of a hypersurface $V(F)$ of degree d in characteristic zero is the statement that the partial derivatives of F have no common zero other than the origin of the affine cone, and Euler's identity $dF = \sum x_i \partial_i F$ shows that such a common zero automatically lies on $V(F)$.

This paper records the parts of this picture that are formalized, in the precise form they are formalized: polynomial identities and an implication about integer points. Everything is stated over $\mathbb{Z}$ with the derivatives entered as explicit polynomials. The repository's `HostLabel` record is a documentation label with string fields; it is not a scheme.

2. Definitions

For $x_0, \ldots, x_5 \in \mathbb{Z}$ define

$$F_5 = \sum_{i=0}^{5} x_i^5, \qquad D_i = 5x_i^4 \ (i = 0, \ldots, 5), \qquad G = F_5 - 5\, x_0x_1x_2x_3x_4x_5.$$

In the formalization, D_i is *defined* to be $5x_i^4$ (Lean names `dF5_dx0`, …, `dF5_dx5`); it is the formal partial derivative $\partial F_5/\partial x_i$, but differentiation itself is not formalized.

3. Results

Theorem 3.1 (Euler identity; `Hodge.VF5.euler_F5`)**.** *For all $x \in \mathbb{Z}^6$, $5F_5(x) = \sum_{i=0}^{5} x_i D_i(x)$.*

Theorem 3.2 (Euler defect of G; `Hodge.VF5.G_euler_defect`)**.** *For all $x \in \mathbb{Z}^6$,*

$$\sum_{i=0}^{5} x_i\Big(5x_i^4 - 5\prod_{j\neq i} x_j\Big) - 5G(x) = -5\prod_{j=0}^{5} x_j.$$

The bracketed factors are the formal partials of G. Since G is not homogeneous (its monomial $\prod x_j$ has degree 6), Euler's identity fails for it, and Theorem 3.2 computes the defect exactly.

Date: November 4, 2026.

Lemma 3.3. (1) *(`Hodge.VF5.partial_as_prod`)* $5x^4 = 5 \cdot x \cdot x \cdot x \cdot x$ *for all* $x \in \mathbb{Z}$.
(2) *(`Hodge.VF5.int_five_mul_prod4_eq_zero`) If* $x \in \mathbb{Z}$ *and* $5 \cdot x \cdot x \cdot x \cdot x = 0$, *then* $x = 0$.

Theorem 3.4 (Isolated critical point of the affine cone; `Hodge.VF5.affine_cone_isolated_at_origin`)**.** *Let* $x \in \mathbb{Z}^6$. *If* $D_i(x) = 0$ *for all* $i = 0, \dots, 5$, *then* $x_0 = x_1 = \cdots = x_5 = 0$.

Proof. For each i, rewrite $D_i(x) = 0$ as $5x_ix_ix_ix_i = 0$ by Lemma 3.3(1); since $\mathbb{Z}$ has no zero divisors, Lemma 3.3(2) gives $x_i = 0$. □

Remark 3.5. Theorem 3.4 is stated for integer points only. The analogous statement over $\mathbb{C}$ (which is what smoothness of $V(F_5)$ requires) follows by the same argument in any integral domain of characteristic $\neq 5$, but that generalization is *not* part of the formalization.

Recorded constants and labels. The following are checked by the kernel but are arithmetic or definitional facts, not geometric computations.

Proposition 3.6. (1) *(`Hodge.VF5.h31_dim`)* $126 - 6 = 120$ *in* $\mathbb{N}$.
(2) *(`Hodge.VF5.sextic_h22_prim_container`)* $1751 = 1751$ *in* $\mathbb{N}$.
(3) *(`Hodge.VF5.sextic_Z_status_uncomputed`) The status string* `sextic_Z_status` *equals* `"uncomputed"`.
(4) *(`Hodge.VF5.host_is_VF5`, `Hodge.VF5.host_dim`) The documentation label has name* `"V(F5)"` *and dimension field* 4.

Item (1) is the subtraction one would use if 126 and 6 were the relevant dimensions in a Hodge-number calculation; item (2) is a reflexivity statement recording a number. Neither the identification of these numbers with Hodge numbers, nor any cycle class, is proved here. Item (3) records that a sextic quantity is explicitly marked as not computed.

4. Scope

Not formalized: schemes, projective space, smoothness as a geometric property, cohomology, Hodge numbers, primitive classes, algebraic cycles, and the Hodge conjecture. The paper claims exactly the integer identities and implications of Theorems 3.1–3.4, Lemma 3.3 and Proposition 3.6.

5. Correspondence between claims and Lean declarations

All declarations are in namespace `Hodge.VF5`, file `Hodge/VF5.lean`, Lean 4 core only.

Claim	Lean declaration
Theorem 3.1	`euler_F5`
Theorem 3.2	`G_euler_defect`
Lemma 3.3(1)	`partial_as_prod`
Lemma 3.3(2)	`int_five_mul_prod4_eq_zero`
Theorem 3.4	`affine_cone_isolated_at_origin`
Proposition 3.6(1)	`h31_dim`
Proposition 3.6(2)	`sextic_h22_prim_container`
Proposition 3.6(3)	`sextic_Z_status_uncomputed`
Proposition 3.6(4)	`host_is_VF5`, `host_dim`

Appendix A. Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/FermatQuinticHost`

- Verified commit:
  `0b56d4b210d9418931af559cc06fdda2ee5b9bd1`
- Lean toolchain: `leanprover/lean4:v4.22.0`
- Mathlib: none (the development depends only on Lean core)
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/FermatQuinticHost/actions/runs/41794839040`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/FermatQuinticHost/archive/0b56d4b210d9418931af559cc06fdda2ee5b9bd1.tar.gz`
  `726065496f5fb2841642e615f7e7c0eb0fd7b3e6927d4bba7dfe4e2dd7d3568a`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 10 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`Hodge.VF5.G_euler_defect`	`[propext, Classical.choice, Quot.sound]`
`Hodge.VF5.affine_cone_isolated_at_origin`	`[propext, Classical.choice, Quot.sound]`
`Hodge.VF5.euler_F5`	`[propext, Classical.choice, Quot.sound]`
`Hodge.VF5.h31_dim`	`(no axioms)`
`Hodge.VF5.host_dim`	`(no axioms)`
`Hodge.VF5.host_is_VF5`	`(no axioms)`
`Hodge.VF5.int_five_mul_prod4_eq_zero`	`[propext]`
`Hodge.VF5.partial_as_prod`	`[propext, Classical.choice, Quot.sound]`
`Hodge.VF5.sextic_Z_status_uncomputed`	`(no axioms)`
`Hodge.VF5.sextic_h22_prim_container`	`(no axioms)`

References

[1] T. Shioda, *The Hodge conjecture for Fermat varieties*, Math. Ann. **245** (1979), 175–184.

[2] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, LNCS 12699 (2021), 625–635.

Email address: `frohmanbenjamin@gmail.com`